

IMechE Design Challenge 2025

Institution of
MECHANICAL
ENGINEERS

Foundation Challenge: Automatic EV Charging Device

S40: Dachi.G, Amit.C, Alberto

Introduction

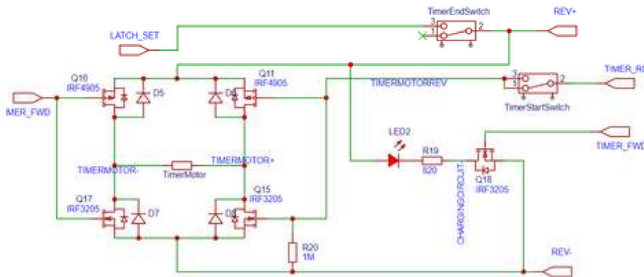
The Design Challenge required us to build a device that simulates an **automatic electric vehicle charging device** using analogue and mechanical principles. It moves forward an unknown distance, makes contact with a vertical target for 15 seconds, and then reverses back to its starting position.

Our solution proposes a device that uses a laser to align with the target before moving forward. It uses features such as 2 rotary timers, a track belt, a 3D printed spiral spring and a DPDT switch to complete its mission.

PCB and Circuitry

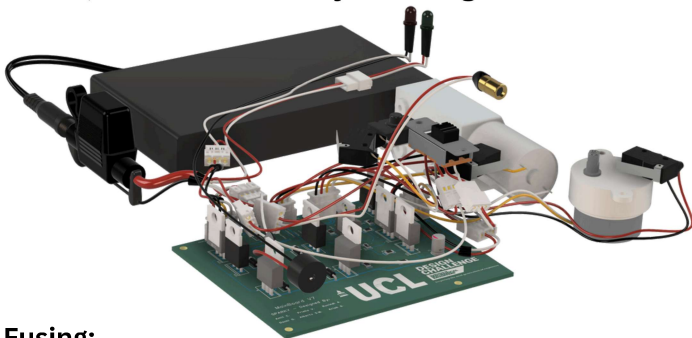
Why PCB?

The PCB forms the core of the electronics system, designed using **EasyEDA**. It provides a robust, compact and reliable platform into which external components can interface with internal ones.



15 second timer system:

4 MOSFETs (2 P-type, 2 N-type) are used in a 'H' configuration to allow for **bidirectional control** of the timer motor. The direction of the rotation is dependent on which P-N pair is active. The active pair is switched by an **SPDT limit switch** at the end of timer, thus **automatically resetting** the timer.

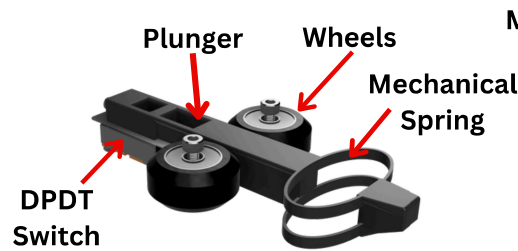


Fusing:

The circuit is fused to 2A. The max **continuous** power draw is 300mA with **instantaneous** spikes up to ~ 1.3A.

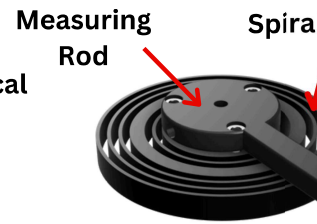
Reverse System:

Design Features



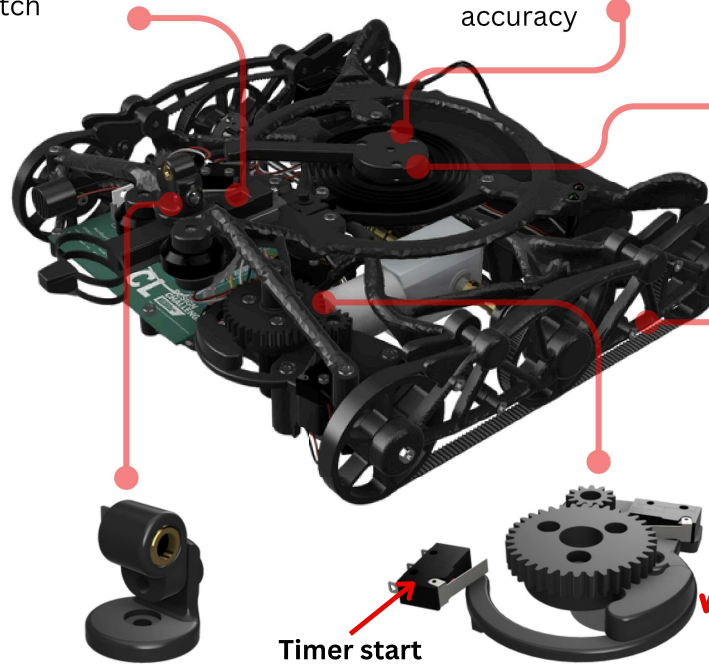
Plunger and DPDT Switch

- Switches between forward and reverse circuits when the device **hits the wall**
- The plunger **directly actuates** the switch
- Plunger integrated with spring to **increase force** on DPDT switch



Spiral Spring (3D Printed with Steel)

- Helps the measuring rod avoid **mechanical binding** with the steel rod
- Keeps the rod **level**
- **Slows down the car** as it approaches the horizontal and vertical target for accuracy



Laser

- **Aligns the device** to maximise the chance of it hitting the vertical target
- Attached to an **adjustable mount** to allow for easy calibration

Timer start switch

Timing Wheel

- The timing wheel is rotated by a separate motor that is geared with such a ratio that it moves from start to end in roughly **seconds**
- The timer **automatically resets back to the starting position** after hitting the switch to improve repeatability and eliminate the need for manual reset

Generative Design